

CHENSHUHAO (CODY) QIN

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EDUCATION

University of Southern California (USC) <i>Master of Science in Computer Science</i> • Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Predictive Modeling	Los Angeles, CA <i>Jan. 2026 – Dec. 2027 (Expected)</i>
Duke University; Duke Kunshan University <i>Bachelor of Science in Computer Science; B.S. in Computation and Design</i> • Honors: Apple WWDC23 Swift Student Challenge Winner (1 of 13 in China), Provincial Innovation Grant Recipient (\$3k), DKU Student Ambassador	Durham, NC / Kunshan, China <i>Aug. 2021 – May. 2025</i>

SKILLS

Programming: Python, TypeScript/JavaScript, Java, SQL (MySQL, PostgreSQL), Swift, C
AI & Machine Learning: PyTorch, TensorFlow, LoRA Fine-tuning, Prompt Engineering, Hugging Face, Pandas, NumPy
LLM & Agents: LLM Agents (AgentScope, LangChain, LlamaIndex), Agent Harness & Context Engineering, Retrieval-Augmented Generation (RAG), Model Serving (vLLM), Embeddings & Vector Search (Milvus), Ray
Infrastructure & Tools: Kubernetes, Docker, CI/CD, Git, Linux, REST APIs (FastAPI, Flask), React, Redis, Object Storage (OSS/S3), AWS, PyTest

WORK EXPERIENCE

Alibaba Cloud (Top-4 Global Cloud Provider) <i>AI Engineer Intern</i> • Collaborated with Alibaba’s Tongyi Lab to productize Data-Juicer Agents — an open-source agent harness — into natural-language-driven multimodal data pipelines (cleaning, filtering, deduplication) for the broader Qwen ecosystem, replacing script-heavy workflows on the Apsara Data Lake platform. • Built an agent-to-Ray distributed execution pipeline over private-cloud object storage (OSS) and Kubernetes — 2,793 records, 16 operators, ~5 minutes — and landed the core capability upstream through a merged PR. • Designed a three-layer context-control architecture (tool-output compaction, memory compression, hard token budgets) cutting single-tool context payloads by 82% and enabling stable multi-turn agents on 30k-token models. • Shipped a full-stack agent workspace (React / FastAPI / MySQL , REST APIs) with persistent sessions, execution tracing, conversation rollback, and multi-instance isolation; delivered 4 production container-image releases . • Contributed 5 merged PRs (PyTest unit coverage, CI green) to the 7k+ Data-Juicer open-source ecosystem across multimodal ingestion, distributed Ray execution, and context optimization; 1 enhancement proposal adopted by maintainers.	Hangzhou, China <i>Jun. 2026 – Aug. 2026</i>
Blueberry AI <i>AI Engineer Intern</i> Built a Milvus + MySQL feature-extraction and auto-tagging pipeline batch-indexing 200K+ images for landmark / IP / SKU retrieval; benchmarked DINOv3, Qwen-VL, and BGE-VL under a unified recall / precision / latency framework for private deployment; delivered a YOLO + Milvus vector search + Gradio retrieval POC validated for commercial use.	Burlingame, CA (Remote) <i>Nov. 2025 – Jan. 2026</i>
Tsinghua Sichuan Energy Internet Research Institute <i>Algorithm Engineer Intern</i> Engineered a DeepLabV3+ (PyTorch) semantic-segmentation model for microgrid fault diagnosis reaching 92% accuracy across complex field scenarios; owned the full data lifecycle of 500+ annotated images from preprocessing to deployment, anchoring winning bids for major energy-infrastructure projects.	Chengdu, China <i>Sep. 2025 – Oct. 2025</i>
Leiting Games (G-bits) <i>AI R&D Intern (AIGC & Automation)</i> Fine-tuned SDXL / Flux LoRA models on Kohya , shipping 30+ ad assets (80% adoption) that drove 117K+ installs and ¥24.6M billings (196% ROI) ; led a 5-person team scaling creative output 3× with 15–20% higher material ROI; built a 98.85%-accurate RPA monitor catching 1,000+ private-server violations (2025 Best Practice).	Shenzhen, China <i>May. 2025 – Aug. 2025</i>

PROJECTS & PUBLICATIONS

ChatDKU — RAG Campus Q&A Assistant <i>Python, LlamaIndex, LangChain, Redis</i> Co-founded a university-adopted Retrieval-Augmented Generation (RAG) Q&A agent indexing 12,000+ webpages over a privately deployed LLM; implemented Hybrid Search (BM25 + vectors) , HyDE , multi-level Redis caching, and SSO web deployment; the system was officially deployed at DKU and recognized by multiple university offices, documented as a DKU Signature Work.	Jul. 2024 – May. 2025
Publication: <i>Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets</i> — ACM SenSys 2024 (Demo Track, second author), pp. 889–890; 95% real-world accuracy via federated learning over BLE.	